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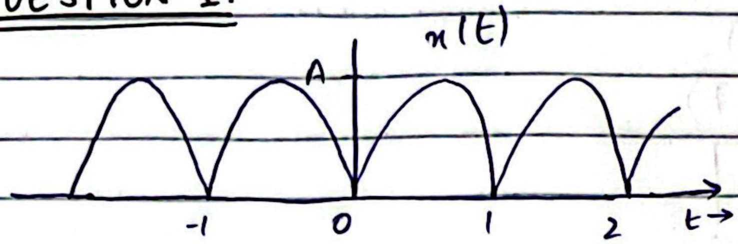
ROLL NO.: 22L-7651

SECTION: 4A(BSEE)

ASSIGNMENT: 3

QUESTION 1:-



sol:-

the formula of compact trigonometric fourier series is

$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n w_0 t + b_n \sin n w_0 t)$$

here

$$x(t) = A \sin \pi t \quad , 0 \le t \le 1$$
$$= -A \sin \pi t \quad , -1 \le t \le 0$$

As given signal has even symmetry

hence $\boxed{b_n = 0}$

now

$$a_n = \frac{2}{T}\int_0^T x(t) \cos n w_0 t \, dt \qquad (\because T = 1) \qquad \left(w_0 = \frac{2\pi}{1} \Rightarrow 2\pi\right)$$

$$a_n = 2\int_0^1 A \sin \pi t \,.\, \cos(2\pi n t)\, dt$$

$$= A\int_0^1 2 \sin \pi t \,.\, \cos(2\pi n t)\, dt$$

$$= A\int_0^1 [\sin(2\pi + 1)nt + \sin(1 - 2\pi)nt]\, dt$$

$$= A\left[\frac{-\cos(2\pi+1)\pi t}{(2n+1)\pi}\right]_0^1 + A\left[\frac{-\cos(1-2n)\pi t}{(1-2n)\pi t}\right]_0^1$$

$$= \frac{-A}{(2n+1)\pi}(-1-1) - \frac{A}{(1-2n)\pi}(-1-1)$$

$$= \frac{2A}{(2n+1)\pi} + \frac{2A}{(1-2n)\pi} \Rightarrow \frac{2A}{\pi}\left(\frac{1}{2n+1} - \frac{1}{2n-1}\right)$$

$$a_n = \frac{2A}{\pi}\left[\frac{2n-1-2n-1}{(2n+1)(2n-1)}\right]$$

$$= \frac{2A}{\pi}\frac{(-2)}{(4n^2-1)}$$

$$\boxed{a_n = \frac{-4A}{\pi(4n^2-1)}}$$

and

$$a_0 = \frac{1}{T}\int_0^T x(t)\,dt$$

$$= 1\int_0^1 A\sin\pi t\,dt$$

$$= \left[\frac{-A\cos\pi t}{\pi}\right]_0^1$$

$$= \frac{-A}{\pi}[-1-1]$$

$$\boxed{a_0 = \frac{2A}{\pi}}$$

Now

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$$

$$x(t) = \frac{2A}{\pi} + \sum_{n=1}^{\infty} \frac{-4A}{\pi(4n^2-1)} \cdot \cos(n\omega_0 t)$$

$a_0 = \frac{2A}{\pi}$, $a_n = \frac{-4A}{\pi(4n^2-1)}$ $\quad \because A_n = \sqrt{a_n^2 + b_n^2}$

$A_n = |a_n|$

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n	1	2	3	4	5
A_n	$\left\|\frac{-4A}{3\pi}\right\|$	$\left\|\frac{-4A}{15\pi}\right\|$	$\left\|\frac{-4\pi}{35\pi}\right\|$	$\frac{-4A}{63\pi}$	...

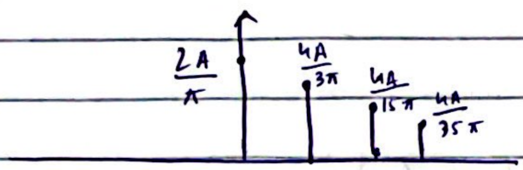
$$x(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n\omega_0 t + \phi_n)$$

$$A_n = \sqrt{a_n^2 + b_n^2} = |a_n|$$

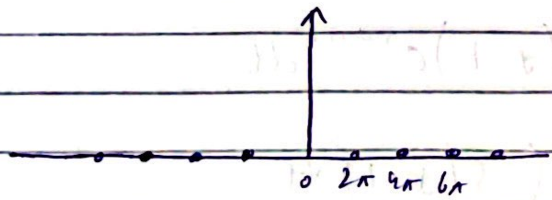
and

$$\phi_n = -\tan^{-1}\frac{b_n}{a_n} \Rightarrow \boxed{\phi_n = 0}$$

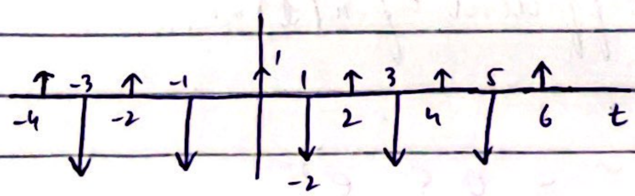
Magnitude spectrum :



Phase spectrum :

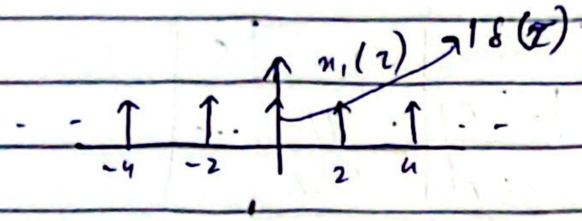


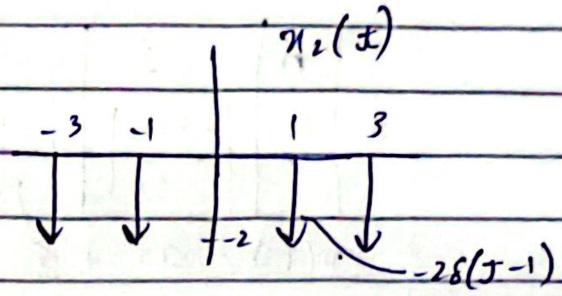
QUESTION 6 :-



sol :-

$$x(t) = x_1(t) + x_2(t)$$





Fourier series coefficient $x_1(t)$ is

$$C_{n_1} = \frac{1}{T_0}\int_0^{T_0} x_1(t)\, e^{-j\omega_0 t n}\, dt = \frac{1}{2}\int_0^{2} \delta(t)\, e^{-j\omega_0 t n}\, dt$$

$$C_{n_1} = \frac{1}{2}\int_0^{2} \delta(t)\, \underbrace{e^{-j\omega_0 \cdot 0 \times n}}_{1}\, dt$$

$$= \frac{1}{2}\int_0^{2} \delta(t)\, dt$$

$$C_{n_1} = \frac{1}{2} \longrightarrow (1)$$

Fourier series coefficient $x_2(t)$ is

$$C_{n_2} = \frac{1}{T_0}\int_0^{T_0} x_2(t)\, e^{-j\omega_0 n t}\, dt$$

$$C_{n_2} = \frac{1}{2}\int_0^{2} -2\delta(t-1)\, e^{-j\omega_0 n t}\, dt$$

$$= \frac{-2e^{-j\omega_0 n}}{2}\int_0^{2} \delta(t-1)\, dt$$

$$C_{n_2} = -e^{-j\omega_0 n} \longrightarrow (2)$$

Fourier series coefficient of $x(t)$ is

$$C_n = C_{n_1} + C_{n_2}$$

from eq (1) & (2)

$$C_n = \frac{1}{2} - e^{-j\omega_0 n} = 0.5 - e^{-j\omega_0 n}$$

$$\omega_0 = \frac{2\pi}{T} \Rightarrow \pi \qquad (\because T = 2)$$

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$$\boxed{C_n = 0.5 - e^{-jn\pi}}$$

the exponential fourier series of $x(t)$ is

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{j\pi n t}$$

$$\boxed{x(t) = \sum_{n=-\infty}^{\infty} \left(0.5 - e^{-jn\pi}\right) e^{jn\pi t}}$$

$$C_n = 0.5 - e^{-jn\pi}$$

$$\cos(n\pi) + j\sin(n\pi) = e^{-jn\pi}$$

$$\sin(n\pi) = 0$$

hence

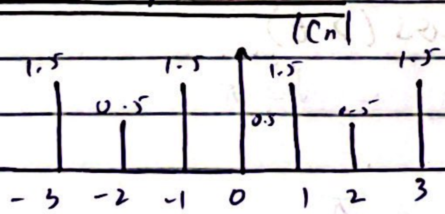
$$e^{-jn\pi} = \cos(n\pi) = (-1)^n$$

$$e^{-j\theta} = \cos\theta - j\sin\theta$$

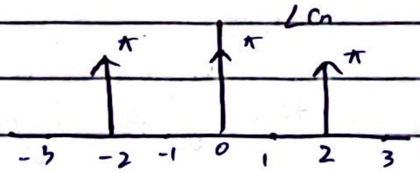
$$C_n = 0.5 - (-1)^n$$

$C_0 = -0.5$	$\lvert C_0 \rvert = 0.5$	$\angle C_0 = \pi$
$C_1 = 1.5$	$\lvert C_1 \rvert = 1.5$	$\angle C_1 = 0$
$C_2 = -0.5$	$\lvert C_2 \rvert = 0.5$	$\angle C_2 = \pi$
$C_{-1} = 1.5$	$\lvert C_{-1} \rvert = 1.5$	$\angle C_{-1} = 0$
$C_{-2} = 0.5$	$\lvert C_{-2} \rvert = 0.5$	$\angle C_{-2} = \pi$

Magnitude spectrum:



Phase spectrum:



QUESTION 7:-

a) $2\sin(3t) + 7\cos(\pi t)$

$$T_1 = \frac{2\pi}{w} = \frac{2\pi}{3}$$

$$T_2 = \frac{2\pi}{w} \Rightarrow \frac{2\cancel{\pi}}{\cancel{\pi}} \Rightarrow 2$$

$$\frac{T_1}{T_2} = \frac{2\pi/3}{2} \Rightarrow \frac{\cancel{2}\pi}{\cancel{6}} \Rightarrow \frac{\pi}{3}$$

As T_1/T_2 is irrational, so the given signal is aperiodic.

b) $7\cos(\pi t) + 5\sin(2\pi t)$

$$T_1 = \frac{2\pi}{w} \Rightarrow \frac{2\pi}{\pi} \Rightarrow 2$$

$$T_2 = \frac{2\pi}{w} \Rightarrow \frac{2\pi}{2\pi} \Rightarrow 1$$

$$\frac{T_1}{T_2} = 2 \qquad \boxed{T_0 = 2}$$

As T_1/T_2 is rational, so the given signal is periodic.

c) $3\cos(\sqrt{2}\,t) + 5\cos(2t)$

$$T_1 = \frac{2\pi}{w} = \frac{2\pi}{\sqrt{2}} = \sqrt{2}\,\pi$$

$$T_2 = \frac{2\pi}{w} = \frac{2\pi}{2} = \pi$$

$$\frac{T_1}{T_2} = \frac{\sqrt{2}\cancel{\pi}}{\cancel{\pi}} \Rightarrow \sqrt{2}$$

As T_1/T_2 is irrational, so the given signal is aperiodic

d) $\sin(5t/2) + 3\cos(6t/2) + 3\sin(t/7 + 30°)$

$T_1 = \frac{2\pi}{w} = \frac{2\pi}{5/2} \Rightarrow \frac{4\pi}{5}$.

$T_2 = \frac{2\pi}{w} = \frac{2\pi}{3\cancel{6}/\cancel{2}} \Rightarrow \frac{2\pi}{3}$

$T_3 = \frac{2\pi}{w} = \frac{2\pi}{1/7} = 14\pi$

$\frac{T_1}{T_2} = \frac{4\pi/5}{2\pi/3} \Rightarrow \frac{\cancel{4}^{2}\pi}{5} \times \frac{3}{2\pi} \Rightarrow \frac{6}{5}$ (rational)

$\frac{T_2}{T_3} = \frac{2\cancel{\pi}}{3} \times \frac{1}{14\cancel{\pi}} \Rightarrow \frac{\cancel{2}\,1}{\cancel{42}\,21} \Rightarrow \frac{1}{21}$ (rational)

So the given signal is periodic

fundamental period $T_0 = LCM\left(\frac{4\pi}{5}, \frac{2\pi}{3}, 14\pi\right)$

$= \frac{LCM(4\pi, 2\pi, 14\pi)}{HCF(5,3,1)}$

$= \frac{28}{1}$

$\boxed{T_0 = 28\pi}$

QUESTION 8:-

$f(t) = 3 + \sqrt{3}\cos 2t + \sin 2t + \sin\left(5t - \frac{\pi}{6}\right) - \frac{1}{2}\left(7t + \frac{\pi}{3}\right)$

a) Consider

$\sqrt{3}\cos 2t + \sin 2t = 2\left(\frac{\sqrt{3}}{2}\cos 2t + \frac{1}{2}\sin 2t\right)$

$= 2\left(\cos\frac{\pi}{6}\cos 2t + \sin\frac{\pi}{6}\sin 2t\right)$

$$= 2\cos(2t - \pi/6)$$

$$\sin\left(5t - \frac{\pi}{6}\right) = \cos\left(5t + \frac{\pi}{3} + \frac{3\pi}{2}\right)$$

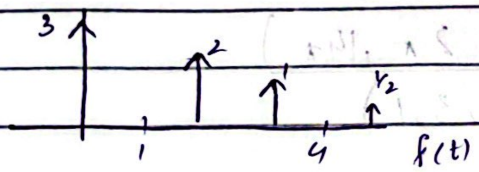
$$= \cos\left(5t + \frac{11}{6}\pi\right)$$

$$-\frac{1}{2}\cos\left(7t + \frac{\pi}{3}\right) = \frac{1}{2}\cos\left(7t + \frac{\pi}{3} + \pi\right)$$

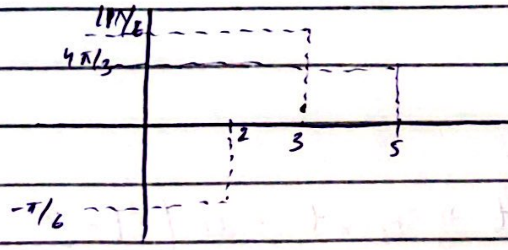
$$= \frac{1}{2}\cos\left(7t + \frac{4}{3}\pi\right)$$

$$\therefore f(t) = 3 + 2\cos(2t - \pi/6) + \cos(5t + {}^{11}/_{6}\pi) + \frac{1}{2}\cos(7t + {}^{4}/_{3}\pi) \quad \rightarrow (1)$$

Amplitude spectra

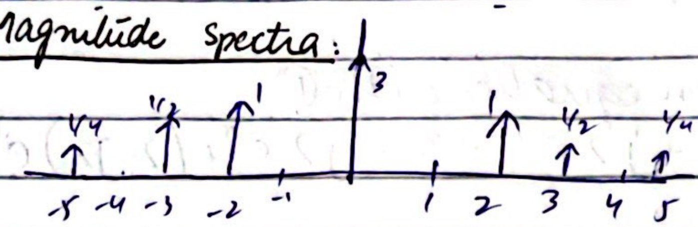


Phase spectra

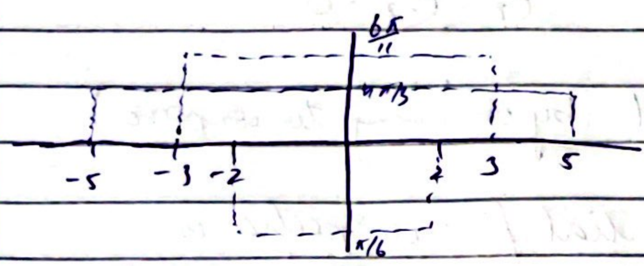


b) We can convert trigonometric exponential fourier series by making half all the trigonometric coefficients except the dc term but phase spectrum remains same since it is two sided

Magnitude spectra:



Phase spectra:



c) the exponential fourier series is

$$f(t) = \frac{e^{j4\pi/3}}{4} e^{-j7t} + \frac{e^{j11/6\pi}}{2} e^{-j5t} + e^{-j\pi/6} e^{-j2t} + 3$$
$$+ e^{-j\pi/6} e^{j2t} + \frac{e^{j11/6}}{2} e^{j5t} + \frac{e^{j4\pi/3}}{4} e^{j5t} \rightarrow (2)$$

d) By using $\cos(x) = \frac{1}{2}[e^{x} + e^{x}]$, we can see that equation (1) & (2) are equal.

<u>QUESTION 9:-</u>

$$x(t) = (2+j2)e^{-j3t} + j2e^{-jt} + 3 - j2e^{jt} + (2-j2)e^{j3t}$$

a) It is known that exponential fourier series is in the form of

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{jn\omega_0 t} \rightarrow ①$$

By comparing given equation with ①

$$f(t) = (2+2j)e^{-j3t} + j2e^{-jt} + 3 - j2e^{jt} + (2-j2)e^{j3t}$$

$C_{-3} = 2+2j$ $\quad$ $C_1 = -j2$

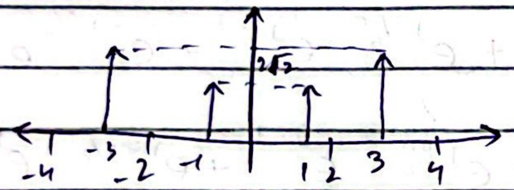
$C_{-1} = j2$ $\quad$ $C_3 = 2-2j$

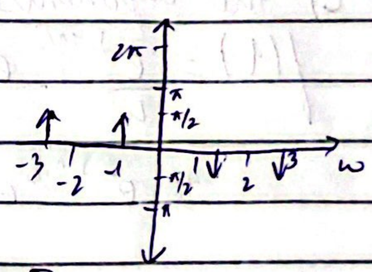
$C_0 = 3$ $\quad$ $C_2 = C_{-2} = 0$

$\omega_0 = 1$ $\quad$ ($\because$ by observing to compare)

the sketch of exponential fourier spectra is

Magnitude spectra:- $\qquad$ **Phase spectra:-**





$$|C_{-3}| = \sqrt{2^2+2^2} = 2\sqrt{2}$$

$$\angle C_{-3} = \tan^{-1}(2/2) = \pi/4$$

b) It is known that trigonometric fourier coefficients are

$$a_0 = C_0$$

$$a_n = (C_n + C_n^*) \text{ or } (C_n + C_{-n})$$

$$b_n = j(C_n - C_n^*) \text{ or } j(C_n - C_{-n})$$

By using above 3 identities

$$a_0 = 3$$

$a_1 = c_1 + c_{-1} = -j2 + j2 = 0$

$a_2 = 0 + 0 = 0$

$a_3 = 2 - 2\cancel{j} + 2 + 2\cancel{j} = 4$

$b_1 = j(-2j - 2j) = -j(4j) = 4$

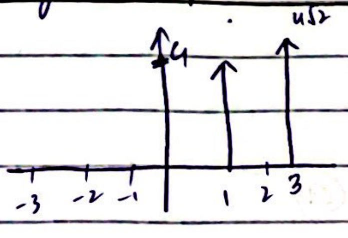
$b_2 = 0$

$b_3 = j(\cancel{2} - 2j - \cancel{2} - 2j) = 4$

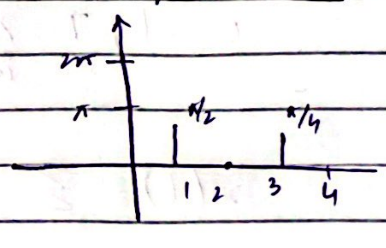
$$f(t) = 3 + 4\cos(3w_0 t) + 4\sin(w_0 t) + 4\sin 3w_0 t$$
$$= 3 + 4\sin w_0 t + 4(\sin 3w_0 t + \cos 3w_0 t)$$
$$= 3 + 4\sin w_0 t + 4 \times \sqrt{2}\left(\frac{1}{\sqrt{2}}\cos 3w_0 t + \frac{1}{\sqrt{2}}\sin 3w_0 t\right)$$
$$= 3 + 4\sin w_0 t + 4\sqrt{2}\left[\cos 3w_0 t - \frac{\pi}{4}\right]$$
$$= 3 + 4\sin t + 4\sqrt{2}(\cos 3t - \pi/4) \quad (\because w_0 = 1)$$

the sketch of ~~exponential~~ trigonometric fourier series is

Magnetic Spectrum :-



Phase spectrum



c) Compact trigonometric series is represented by

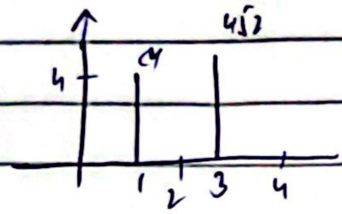
$$f(t) = \sum_{n=0}^{\infty} r_n \cos(n w_0 t - \phi_n)$$

$$r_n = \sqrt{a_n^2 + b_n^2}$$

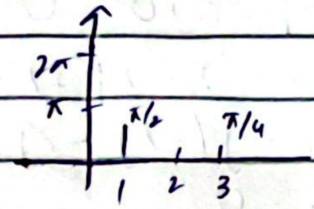
$$\phi_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$f(t) = 3 + 4\cos\left(t - \frac{\pi}{2}\right) + 4\sqrt{2}\cos\left(3t - \frac{\pi}{4}\right)$$

Magnitic Spectrum

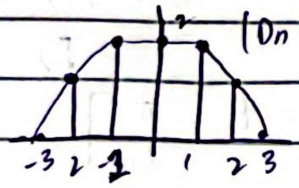


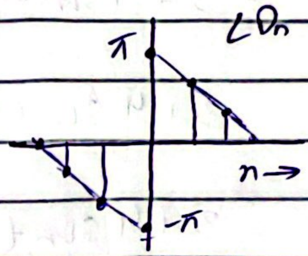
Phase spectrum



d) Bandwidth $= \omega_0 =$

QUESTION 10 :-





a) Exponential fourier series $= \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$

Here

$\omega_0 = 1, \; T = \frac{2\pi}{\omega_0} \Rightarrow 2\pi$

So

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jnt} \longrightarrow (a)$$

the trigonometric fourier series is given by

$$x(t) = \sum_{n=1}^{\infty} \left[a_n \cos n\omega_0 t + b_n \sin n\omega_0 t\right] + a_0$$

$$x(t) = \sum_{n=0}^{\infty} r_n \cos(n\omega_0 t - \phi_n)$$

where $r_n = \sqrt{a_n^2 + b_n^2}$ and $\phi_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$

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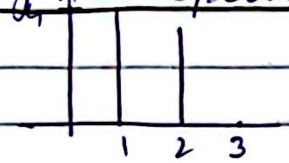
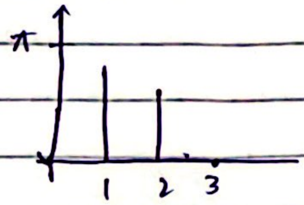
If we calculate the relation between exponential fourier series coefficient and the trigonometric fourier series coefficient, then it comes out to be

$$\frac{r_n}{2} = |D_n| \quad \text{and} \quad \angle D_n = -\phi_n$$

b) the trigonometric fourier series spectra is

Magnetic Spectra :- Phase Spectra :-





c) the trigonometric fourier series for spectra in part (b) is

$$x(t) = \sum_{n=0}^{\infty} r_n \cos(nt - \phi_n)$$

d) We know that D_n is given by

$$D_n = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$D_n = \underbrace{\frac{1}{T}\int_{-T/2}^{T/2} x(t)(\cos n\omega_0 t)\,dt}_{a_n/2} - j\underbrace{\frac{1}{T}\int_{-T/2}^{T/2} x(t)\sin n\omega_0 t\, dt}_{b_n/2}$$

$$D_n = \frac{a_n}{2} - j\frac{b_n}{2} \qquad \angle D_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$|D_n| = \frac{\sqrt{a_n^2 + b_n^2}}{2} = \frac{r_n}{2} \qquad \angle D_n = -\phi_n$$

Fourier series defined either in real form or complex form are same and they are unique for a signal $x(t)$.

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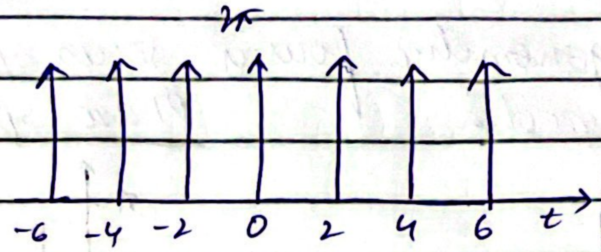
complex form are same and they are unique for a signal $x(t)$.

QUESTION 5:-



the given periodic signal is an impulse train with amplitude 2π and the period $T=2$

Lets define the signal for one period:

$x(t) = 2\pi\delta(t)$

the compact trigonometric form is

$x(t) = a_0 + \sum_{n=1}^{\infty} A_n \cos(n w_0 t - \theta_n)$

$w_0 = \frac{2\pi}{T} = \pi$

$\int_0^2 \delta(t)\, dt = 1$

$a_0 = \frac{1}{T}\int_0^T x(t)\, dt = \frac{1}{2}\int_0^2 2\pi\, \delta(t)\, dt$

$= \cancel{2}\frac{\pi}{\cancel{2}}(1)$

$= \pi$

$a_n = \frac{2}{T}\int_0^T x(t) \cos(n w_0 t)\, dt$

$= \frac{2}{2}\int_0^2 2\pi\, \delta(t) \cos(n w_0 t)\, dt$

$$= 2\pi \int_0^2 \delta(t) \cos(0)\, dt$$

$$= 2\pi$$

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t)\, dt$$

$$= \frac{2}{2} \int_0^2 2\pi\, \delta(t) \sin(n\omega_0 t)\, dt$$

$$= 2\pi \int_0^2 \delta(t) \sin(0)\, dt$$

$$= 0$$

$$A_n = \sqrt{a_n^2 + b_n^2}$$

$$= 2\pi$$

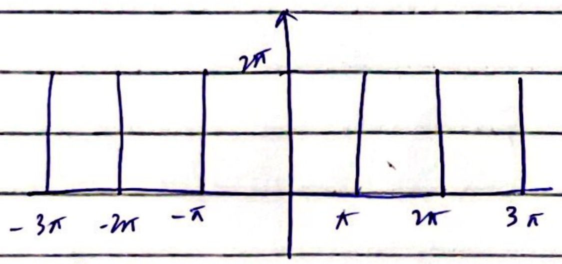
$$\theta_n = \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$= 0$$

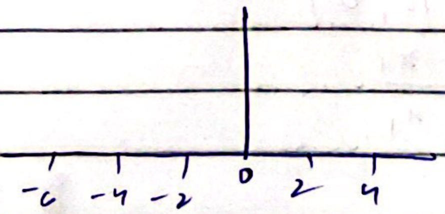
hence

$$x(t) = \pi + \sum_{n=1}^{\infty} 2\pi \cos(\pi n t)$$

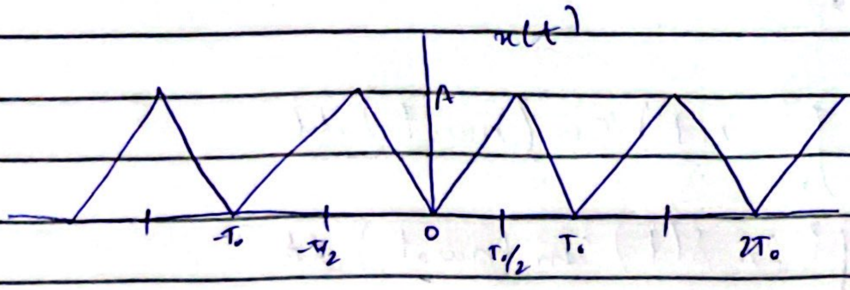
magnitude spectrum :-



Phase spectraum :-

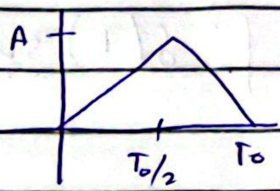


QUESTION 2:-



$T_0 = T_0$

$$\omega_0 = \frac{2\pi}{T_0}$$



$$\omega_0 = \frac{2\pi}{T_0}$$

Equation of straight line :

$(0,0)$ to $(T_0/2, A)$

$$\frac{y_2 - y_1}{y - y_1} = \frac{x_2 - x_1}{x - x_1}$$

$$\frac{A - 0}{y - 0} = \frac{T_0/2 - 0}{x - 0}$$

$$\frac{A}{y} = \frac{T_0/2}{x}$$

$$Ax = y\frac{T_0}{2}$$

$$y = \frac{2Ax}{T_0}$$

$$y = \frac{2At}{T_0}$$

$(T_0/2, A)$ to $(T_0, 0)$

$$\frac{y_2 - y_1}{y - y_1} = \frac{x_2 - x_1}{x - x_1}$$

$$\frac{0-A}{y-A} = \frac{T_0 - T_0/2}{x - T_0/2}$$

$$\frac{-A}{y-A} = \frac{2T_0 - T_0/\cancel{2}}{2x - T_0/\cancel{2}}$$

$$\frac{-A}{y-A} = \frac{T_0}{2x - T_0}$$

$$-A(2x - T_0) = T_0(y-A)$$

$$\frac{-2Ax + AT_0}{T_0} = y - A$$

$$\frac{-2Ax}{T_0} + A + A = y$$

$$y = \frac{-2Ax}{T_0} + 2A$$

$$y = 2A\left(\frac{-x}{T_0} + 1\right) \rightarrow 2A\left(1 - \frac{t}{T_0}\right)$$

$$y = \begin{cases} 2A\, t/T_0 & , 0 < t < T_0/2 \\ 2A(1 - t/T_0) & , T_0/2 < t < T_0 \end{cases}$$

$$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt$$

$$a_0 = \frac{1}{T_0}\left[\int_0^{T_0/2} \frac{2At}{T_0}\,dt + \int_{T_0/2}^{T_0} 2A\left(1 - \frac{t}{T_0}\right)dt\right]$$

$$= \frac{1}{T_0}\left[\frac{1}{T_0}(\cancel{2}A)\left(\frac{t^2}{\cancel{2}}\right)\Big|_0^{T_0/2}\right) + 2A\left(\int_{T_0/2}^{T_0} 1\,dt - \frac{1}{T_0}\int_{T_0/2}^{T_0} t\,dt\right]$$

$$= \frac{1}{T_0}\left[\frac{A}{\cancel{T_0}}\left(\frac{T_0^{\cancel{2}}}{4} - 0\right) + 2A\left(t\Big|_{T_0/2}^{T_0} - \frac{1}{T_0}\left(\frac{t^2}{2}\right)\Big|_{T_0/2}^{T_0}\right]$$

$$= \frac{1}{T_0}\left[\frac{AT_0}{4} + 2A\left[\left(T_0 - \frac{T_0}{2}\right) - \frac{1}{2T_0}\left(T_0^2 - \frac{T_0^2}{4}\right)\right]\right]$$

Scanned with CamScanner

$$a_0 = \frac{1}{T_0}\left[\frac{AT_0}{4} + 2A\left(\frac{T_0}{2} - \frac{1}{2\cancel{T_0}}\left(\frac{3T_0^{\cancel{2}}}{4}\right)\right)\right]$$

$$a_0 = \frac{1}{T_0}\left[\frac{AT_0}{4} + 2A\left(\frac{T_0}{2} - \frac{3T_0}{8}\right)\right]$$

$$= \frac{1}{T_0}\left[\frac{AT_0}{4} + \cancel{2}A\left(\frac{8T_0 - 6T_0}{\cancel{16}\,8}\right)\right]$$

$$= \frac{1}{\cancel{T_0}}\left[\frac{\cancel{2}A\cancel{T_0}}{\cancel{4}\,2}\right]$$

$$a_0 = \frac{A}{2}$$

$$a_n = \frac{2}{T_0}\int_0^{T_0} x(t)\cos n w_0 t\, dt$$

$$= \frac{2}{T_0}\left[\int_0^{T_0} x(t)\cos n\frac{2\pi}{T_0}t\, dt\right]$$

$$a_n = \frac{2}{T_0}\left[\underbrace{\int_0^{T_0/2}\frac{2At}{T_0}\cos n\frac{2\pi}{T_0}t\, dt}_{1} + \underbrace{\int_{T_0/2}^{T_0} 2A\left(1-\frac{t}{T_0}\right)\cos n\frac{2\pi t}{T_0}\, dt}_{2}\right]$$

from eq 1

$$\int_0^{T_0/2}\frac{2At}{T_0}\cos n\frac{2\pi}{T_0}t\, dt$$

$$= \frac{2A}{T_0}\left[t\int\cos n\frac{2\pi}{T_0}t\, dt - \left[\int\cos n\frac{2\pi}{T_0}\frac{d}{dt}(t)\right]dt\right]$$

$$= \frac{2A}{T_0}\left[\frac{t\sin n2\pi/T_0\, t}{2n\pi/T_0} - \int\left[\frac{\sin n2\pi/T_0}{2n\pi/T_0}\right]dt\right]$$

$$= \frac{2A}{T_0}\left[\frac{tT_0}{2n\pi}\sin n\frac{2\pi}{T_0}t + \frac{T_0^2}{4\pi^2 n^2}\cos n\frac{2\pi}{T_0}t\right]$$

Applying the limits $0 \rightarrow T_0/2$

$$= \frac{2A}{T_0}\left[\frac{t\,T_0}{2n\pi}\sin\frac{2\pi}{T_0}t + \frac{T_0^2}{4n^2\pi^2}\cos n\frac{2\pi t}{T_0}\right]$$

$$= 2A\left[\frac{T_0}{2}\,\frac{T_0}{2\pi n}\sin n\frac{\cancel{2}\pi}{\cancel{T_0}}\times\frac{\cancel{T_0}}{\cancel{2}} + \frac{T_0^2}{4n^2\pi^2}\cos n\frac{\cancel{2}\pi}{\cancel{T_0}}\times\frac{\cancel{T_0}}{\cancel{2}}\right]$$

$$- \left(0\times\frac{T_0}{2n\pi}\times 0 + \frac{T_0^2}{4n^2\pi^2}\times 1\right)$$

$$= \frac{2A}{T_0}\left[\frac{T_0^2}{4n\pi}\sin n\pi + \frac{T_0^2}{4n^2\pi^2}\cos n\pi - \frac{T_0^2}{4n^2\pi^2}\right]$$

Now $\sin \pi n = 0$, n is any integer

$$= \frac{2A}{T_0}\left[\frac{T_0^3}{4n\pi}\times 0 + \frac{T_0^2}{4n^2\pi^2}\cos n\pi - \frac{T_0^2}{4n^2\pi^2}\right]$$

$$= \frac{2A}{\cancel{T_0}}\times\frac{T_0^{\cancel{2}}}{4n^2\pi^2}\left[-1+\cos n\pi\right]$$

$$= \frac{2A\,T_0}{4n^2\pi^2}\left[\cos n\pi - 1\right] \longrightarrow (a)$$

from eq ②

$$\int_{T_0/2}^{T_0} 2A\left(1-\frac{t}{T_0}\right)\cos n\frac{2\pi}{T_0}t\,dt$$

$$= 2A\left[\int \cos n\frac{2\pi}{T_0}t\,dt - \frac{1}{T_0}\int t\cos n\frac{2\pi}{T_0}t\,dt\right]$$

$$= 2A\left[\frac{\sin n 2\pi t/\cancel{T_0}}{2n\pi/\cancel{T_0}} - \frac{1}{T_0}\left(\int t\cos n\frac{2\pi}{T_0}t\,dt\right)\right]$$

$$= 2A\left[\frac{T_0}{2\pi n}\sin n\frac{2\pi}{T_0}t - \frac{1}{T_0}\left(t\,\frac{\sin n\frac{2\pi}{T_0}}{2n\pi/T_0} - \int\left(\frac{\sin\frac{n2\pi t}{T_0}\times 1}{n\frac{2\pi}{T_0}}\right)dt\right)\right]$$

$$= 2A\left[\frac{T_0}{2n\pi}\sin\frac{n2\pi t}{T_0} - \frac{1}{T_0}\left(\frac{t\,T_0}{2n\pi}\sin\frac{n2\pi t}{T_0} - \frac{T_0}{2n\pi}\int\sin\frac{2\pi}{T_0}t\,dt\right)\right]$$

$$= 2A\left[\frac{T_0}{2n\pi}\sin\frac{2\pi}{T_0}t - \frac{1}{T_0}\left(\frac{t\,T_0}{2n\pi}\sin\frac{2\pi n t}{T_0} + \frac{T_0}{2n\pi}\times\frac{T_0}{2n\pi}\cos\frac{n2\pi}{T_0}\right)\right]$$

$$= 2A\left[\frac{T_0}{2n\pi}\sin n\frac{2\pi}{T_0}t - \frac{1}{T_0}\left(\frac{t\,t_0}{2n\pi}\sin\frac{2\pi nt}{T_0} + \frac{T_0^2}{4n^2\pi^2}\left\{\cos n\frac{2\pi}{T_0}t\right)\right)\right]$$

$$= 2A\left[\frac{T_0}{2n\pi}\sin n\frac{2\pi}{T_0}t - \frac{t}{2n\pi}\sin\frac{2\pi t}{T_0} - \frac{T_0}{4n^2\pi^2}\cos n\frac{2\pi t}{T_0}\right]$$

Applying the limit

$$= 2A\left[\frac{T_0}{2n\pi}\sin\frac{2n\pi t}{T_0} - \frac{t}{2n\pi}\sin\frac{2\pi t}{T_0} - \frac{T_0}{4n^2\pi^2}\cos\frac{n2\pi t}{T_0}\right]$$

$$= 2A\left[\left(\frac{T_0}{2n\pi}\sin\frac{2n\pi}{\cancel{T_0}}\times\cancel{T_0} - \frac{T_0}{2n\pi}\sin\frac{2\pi n}{\cancel{T_0}}\times\cancel{T_0} - \frac{T_0}{4n^2\pi^2}\cos\frac{2n\pi}{\cancel{T_0}}\times\cancel{T_0}\right)\right.$$

$$\left. - \left(\frac{T_0}{2n\pi}\sin\frac{\cancel{2}n\pi}{\cancel{T_0}}\times\frac{\cancel{T_0}}{\cancel{2}} - \frac{T_0/2}{2n\pi}\sin\frac{\cancel{2}\pi n}{\cancel{T_0}}\times\frac{\cancel{T_0}}{\cancel{2}} - \frac{T_0}{4n^2\pi^2}\cos\frac{\cancel{2}n\pi}{\cancel{T_0}}\times\frac{\cancel{T_0}}{\cancel{2}}\right)\right]$$

$$= 2A\left[\left(\frac{T_0}{2n\pi}\sin 2n\pi - \frac{T_0}{2n\pi}\sin 2\pi n - \frac{T_0}{4n^2\pi^2}\cos 2n\pi\right) - \left(\frac{T_0}{2n\pi}\sin n\pi - \frac{T_0}{4n\pi}\sin n\pi - \frac{T_0}{4n^2\pi^2}\cos n\pi\right)\right]$$

$$= 2A\left[\frac{-T_0}{4n^2\pi^2}\cos 2n\pi - \left(0 - 0 - \frac{T_0}{4n^2\pi^2}\cos n\pi\right)\right]$$

$$= 2A\left[\frac{-T_0}{4n^2\pi^2} + \frac{T_0}{4n^2\pi^2}\cos n\pi\right]$$

$$= 2A\left[\frac{T_0}{4n^2\pi^2}\right](-1 + \cos n\pi) \rightarrow (b)$$

From (a) and (b), the eq (1) becomes

$$a_n = \frac{2}{T_0}\left[\int_0^{T_0/2}\frac{2At}{T_0}\cos\frac{n2\pi t}{T_0}\,dt + \int_{T_0/2}^{T_0} 2A\left(1 - \frac{t}{T_0}\right)\cos\frac{n2\pi t}{T_0}\,dt\right]$$

$$= \frac{2}{T_0}\left[\frac{2A\,T_0}{4n^2\pi^2}[\cos n\pi - 1] + 2A\frac{T_0}{4n^2\pi^2}(\cos n\pi - 1)\right]$$

$$a_n = \frac{\cancel{2}}{\cancel{T_0}}\times\frac{2A\cancel{T_0}}{\cancel{4}n^2\pi^2}\left[-2 + 2\cos n\pi\right]$$

$$a_n = \frac{A}{n^2\pi^2}\left[-2 + 2\cos n\pi\right]$$

$$a_n = \frac{2A}{n^2\pi^2}\left[-1 + \cos n\pi\right]$$

As ~~Ea~~ it is an even function, so

$b_n = 0$

therefore

the compact trigonometric fourier series is

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left(a_n \cos n w_0 t\right)$$

$$x(t) = \frac{A}{2} + \sum_{n=1}^{\infty}\left(\frac{2A}{n^2\pi^2}\left(-1 + \cos n\pi\right)\cos n\frac{2\pi}{T_0}t\right]$$

$$x(t) = \frac{A}{2} + \frac{2A}{\pi^2}\left[-\frac{2}{1}\cos\frac{2\pi t}{T_0} - \frac{2}{5^2}\cos\frac{10\pi}{T_0} - \frac{2}{3^2}\cos\frac{6\pi t}{T_0} - \frac{2}{7^2}\cos\frac{14\pi t}{T_0}\right.$$

$$x(t) = \frac{A}{2} - \frac{4A}{\pi^2}\cos\frac{2\pi}{T_0} - \frac{4A}{25\pi^2}\cos\frac{10\pi t}{T_0} - \frac{4A}{9\pi^2}\cos\frac{6\pi t}{T_0} - \frac{4A}{49\pi^2}\cos\frac{14\pi t}{T_0} \cdots$$

For compact trigonometric fourier series, All terms must be cosine terms with +ve amplitudes.

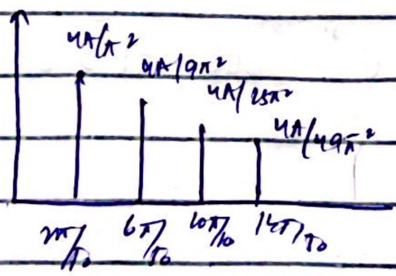
$-\cos\theta = \cos(\theta - \pi)$

therefore

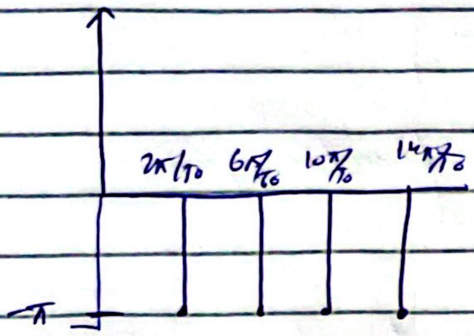
$$x(t) = \frac{A}{2} + \frac{4A}{\pi^2}\cos\left(\frac{2\pi t}{T_0} - \pi\right) + \frac{4A}{9\pi^2}\cos\left(\frac{6\pi t}{T_0} - \pi\right)$$

$$+ \frac{4A}{25\pi^2}\cos\left(\frac{10\pi t}{T_0} - \pi\right) + \frac{4A}{49\pi^2}\cos\left(\frac{14\pi t}{T_0} - \pi\right) + \cdots$$

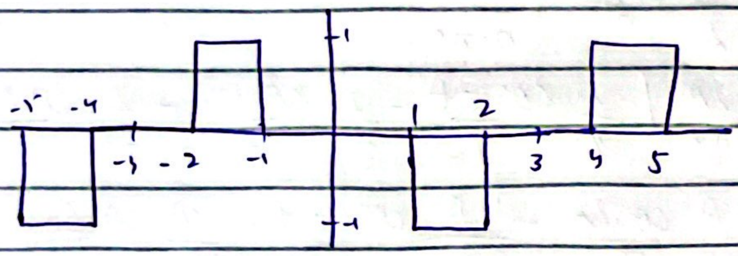
Amplitude Spectrum :-



Phase Spectrum :-



QUESTION 3 :-



$T_0 = 6$

$\omega_0 = \dfrac{2\pi}{T_0}$

$\omega_0 = \dfrac{\pi}{3}$

As

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

$$D_n = \frac{1}{T_0}\int_{-2}^{2} x(t)\, e^{-jn\omega_0 t}\, dt$$

$$D_n = \frac{1}{6}\left[\int_{-2}^{-1} 1\, e^{-jn\pi/3 t}\, dt + \int_{-1}^{0} 0\, e^{-jn\pi/3 t}\, dt + \int_{1}^{2} -1\, e^{-jn2\pi/3\, t}\, dt\right]$$

$$= \frac{1}{6}\left[\left.\left(\frac{e^{-jn\pi/3 t}}{-jn\pi/3}\right)\right|_{-2}^{-1} + 0 - \left.\frac{e^{-jn\pi/3}}{-jn\pi/3}\right|_{1}^{2}\right]$$

$$D_n = \frac{1}{6}\left[\frac{-3}{jn\pi}\left(e^{jn\pi/3} - e^{j2n\pi/3}\right) + \left(\frac{3}{jn\pi}\left(e^{-j2n\pi/3} - e^{-jn\pi/3}\right)\right)\right]$$

$$D_n = \frac{1}{2\cancel{6}} \times \frac{\cancel{3}}{jn\pi}\left[-e^{jn\pi/3} + e^{j2n\pi/3} + e^{-j2n\pi/3} - e^{-jn\pi/3}\right]$$

$$D_n = \frac{1}{jn\pi}\left(-\cos n\frac{\pi}{3} + \cos 2n\frac{\pi}{3}\right)$$

therefore, the exponential fourier series is

$$x(t) = \sum_{n=-\infty}^{\infty} D_n e^{jn\omega_0 t}$$

$$= \sum_{n=-\infty} \frac{1}{jn\pi}\left[\cos 2n\frac{\pi}{3} - \cos n\frac{\pi}{3}\right] e^{jn\pi/3\, t}$$

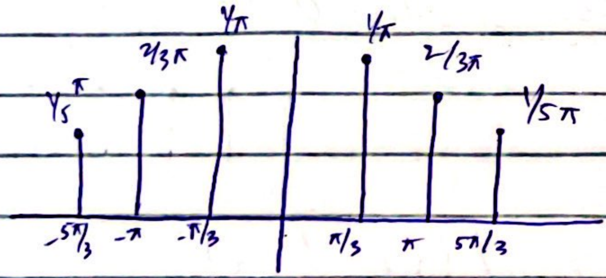
$$= \frac{1}{j\pi}\sum_{n=-\infty}^{\infty}\left[\frac{1}{n}\left(\cos 2n\frac{\pi}{3} - \cos n\frac{\pi}{3}\right) e^{jn\pi/3t}\right]$$

$$= \frac{1}{j\pi}\left[\frac{-1}{1}e^{j\pi/3t} + \frac{2}{3}e^{j3\pi/3t} - \frac{1}{5}e^{j5/3\pi t} - \frac{1}{-1}e^{-j\pi/3t} + \frac{2}{-3}e^{-j3/3\pi t} \cdots\right]$$

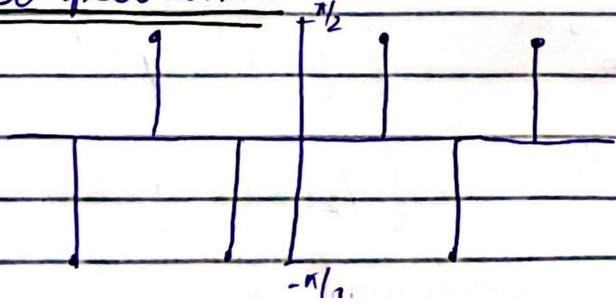
$$= -\frac{1}{j\pi}e^{j\pi/3t} + \frac{2}{j3\pi}e^{j\pi t} - \frac{1}{j5\pi}e^{j5/3\pi t} + \frac{1}{j\pi}e^{-j\pi/3t} - \frac{2}{3j\pi}e^{-j\pi t} + \cdots]$$

$$x(t) = (0.3)e^{j\pi/2}e^{j\pi/3t} + 0.0636e^{j\pi/2}e^{j5/3\pi t} + 0.212e^{j\pi/2}e^{j\pi t} + 0.310e^{-j\pi/2}e^{-}$$

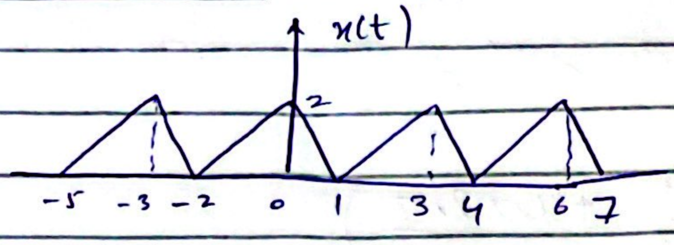
Amplitude Spectrum:



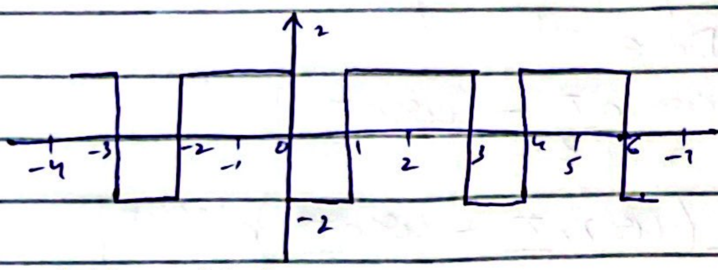
Phase Spectrum:



QUESTION 4 :-



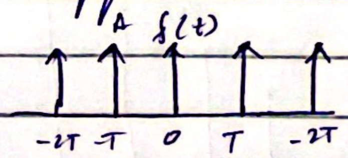
differentiate at $x(t)$



Again differentiate :-

Let $x(t)$ have fourier series coefficient (C_n)
periodic impulse fourier series coefficient

$$C_n = \frac{A}{T}$$



$$f_3(t) = \frac{d^2 x(t)}{dt^2} = f_1(t) - f_2(t)$$

$$f_1(t) \longleftrightarrow C_{n_0} e^{-jn\omega_0 \times 1} \qquad \{ f_1(t) = f(t-1)$$

$$f_2(t) \longleftrightarrow C_{n_1} e^{jn\omega_0 (0)} \qquad \{ f_2(t) = f(t)$$

$f_2(t) \longleftrightarrow Cn_1$

Fourier series coefficient of $x(t)$

Cn_2

$$(jn\omega_0)^2 Cn_2 = \frac{3}{3} e^{-jn\omega_0} - \frac{3}{3}$$

$$\left(jn\frac{2\pi}{3}\right)^2 Cn_2 = e^{-jn2\pi/3} - 1$$

$$Cn_2 = \frac{e^{-jn2\pi/3} - 1}{(jn2\pi/3)^2}$$

for $n = 1, 2, 3, \ldots$

$$Cn_0 = \frac{1}{3}\int_{-2}^{1} x(t)\,dt$$

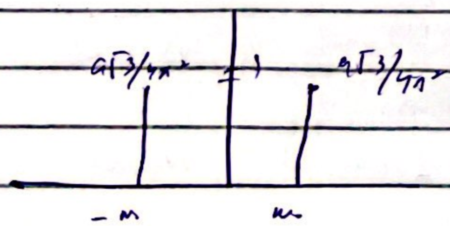
$$= \frac{1}{3}\int_{-2}^{1} x(t)\,dt$$

$$= \frac{1}{3}\left[\frac{1}{2} \times -2 \times 2 + \frac{1}{2} \times 1 \times 2\right]$$

$$= \frac{1}{2}\left[2+1\right]$$

$Cn_0 = 1$

Amplitude Spectrum:



$$|C_{n_1}| = \left| \frac{e^{-j2\pi/3} - 1}{(j2\pi/3)^2} \right|$$

$$= \left| \frac{-1/2 - \sqrt{3}/2\, j - 1}{4\pi^2/9} \right|$$

$$|C_{n_1}| = \frac{\sqrt{9/4 + 3/4}}{4\pi^2/9}$$

$$= \frac{9\sqrt{3}}{4\pi^2}$$

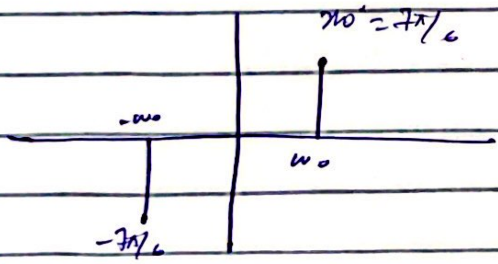
$$\angle C_{n_1} = 180 + \tan^{-1} \frac{\sqrt{3}/2}{3/2}$$

$$= 180 + \tan^{-1} \frac{1}{\sqrt{3}}$$

$$= 180 + 30$$

$$= 210^\circ$$

<u>Phase Spectrum:</u>



Hence fourier series coefficient of given $x(t)$ is

$$C_{n_2} = \begin{cases} \dfrac{e^{-jn2\pi/3} - 1}{(jn2\pi/3)^2} & n \neq 0 \\ 1 & n = 0 \end{cases}$$